ABSTRACT

Deep Hybrid Framework for Paddy Disease Detection and Remedial Recommendation Using MLOA Optimization

Paddy diseases such as Blast, Bacterial Leaf Blight, and Hispa pose a severe threat to global food security, causing approximately 20% of annual rice yield loss. Traditional disease identification methods rely heavily on manual visual inspection or basic image processing techniques that utilize hand-crafted statistical features. While effective to a degree, these existing methods often fail to capture complex visual patterns in early-stage infections and lack actionable post-diagnosis support, limiting their practical utility for farmers.

This project proposes a Deep Hybrid Framework that integrates Transfer Learning with Swarm Intelligence to enhance the accuracy and efficiency of paddy disease detection. Unlike conventional approaches that depend on manual feature extraction, the proposed system utilizes a pre-trained Convolutional Neural Network (CNN), such as MobileNetV3, to automatically extract high-dimensional deep features from paddy leaf images. To address the computational complexity and noise associated with deep features, a Modified Lemurs Optimization Algorithm (MLOA) is implemented as a feature selection layer. This "Deep Hybrid" approach optimizes the feature vector before classification, ensuring superior accuracy compared to standard statistical methods.

Furthermore, to transform the system from a passive diagnostic tool into an active disease management solution, a Remedial Recommendation Module is integrated. Upon detecting a disease, the system queries a specialized agricultural knowledge base to provide instant, specific prescriptions for chemical pesticides and organic alternatives. The proposed framework demonstrates a robust "Diagnosis-to-Cure" workflow, offering a scalable, user-friendly, and high-precision solution for modern smart agriculture.

1. Statement About the Problem

- **Economic Impact:** Pests and diseases are major issues in paddy production, causing farmers to lose approximately 20% of rice yield worldwide.
- **Limitations of Manual Inspection:** Traditional disease identification relies on visual inspection of crops, which is subjective, labor-intensive, and prone to human error, often leading to late diagnosis.
- **Inadequacy of Existing Tech:** Current automated methods, as seen in the base research, rely on manual "Statistical Feature Extraction" (calculating Mean, Variance, Entropy, etc.) from thermal images. These techniques often fail to

capture complex visual patterns like texture variations and lesion borders, limiting accuracy to around 90%.

- **Lack of Remedial Action:** Most existing systems focus solely on detection (identifying the disease) but fail to provide actionable remedial advice (pesticides or cures), leaving the farmer without an immediate solution.

2. Scope

- **Hybrid Intelligence:** The scope of this research encompasses the development of a "Deep Hybrid" system that combines **Deep Learning (Transfer Learning)** for robust feature extraction with **Swarm Intelligence (MLOA)** for feature optimization.
- **Advanced Feature Extraction:** Unlike the base paper which used 14 statistical features, this project uses pre-trained Convolutional Neural Networks (CNNs) like **MobileNetV3** to extract high-dimensional deep features from paddy leaf images.
- **Optimization:** It implements the **Modified Lemurs Optimization Algorithm (MLOA)** to select the most discriminative features from the deep learning vector.
- **Remedial Recommendation:** The system integrates a knowledge base to provide specific chemical and organic treatment suggestions based on the detected disease.
- **Deployment:** The project aims to provide a user-friendly interface for non-expert users (farmers) to upload images and receive instant diagnosis and care instructions.

3. Existing Method

- **Data Source:** The existing method relies primarily on **Thermal Images** of paddy leaves (captured using FLIR cameras).
- **Feature Engineering:** It utilizes a "Filter-Based Feature Transformation" technique where **14 statistical features** (7 original + 7 Box-Cox transformed) are manually extracted from each image. These include Mean, Variance, Entropy, Skewness, Kurtosis, Variation, and Standard Error of Mean.
- **Algorithm:** It employs a **Modified Lemurs Optimization Algorithm (MLOA)**, inspired by the Sine Cosine Algorithm, to transform these statistical features into a linearly separable format.
- **Classification:** It uses traditional supervised classifiers like **K-Nearest Neighbor (KNN)**, Random Forest (RFC), and Linear Discriminant Analysis (LDA) to predict the disease class.

4. Disadvantages of Existing System

1. **Limited Accuracy:** The reliance on simple mathematical features (like Mean/Variance) limits the model's ability to distinguish between visually similar diseases, capping accuracy at roughly 90%.
2. **Manual Feature Engineering:** "Hand-crafting" features is time-consuming and may miss critical visual cues that Deep Learning models automatically detect.
3. **Hardware Dependency:** The existing method relies heavily on **Thermal Cameras**, which are expensive and less accessible to common farmers compared to standard smartphone cameras (RGB).
4. **No Actionable Advice:** The system identifies the disease but does not suggest any pesticides or cures, failing to close the loop for the farmer.
5. **Complexity:** The base paper admits that while Deep Learning is accurate, it is often avoided due to complexity; however, using manual features trades off potential accuracy for simplicity.

5. Proposed System

The proposed system introduces a **Deep Hybrid Framework** to overcome the limitations of the existing method.

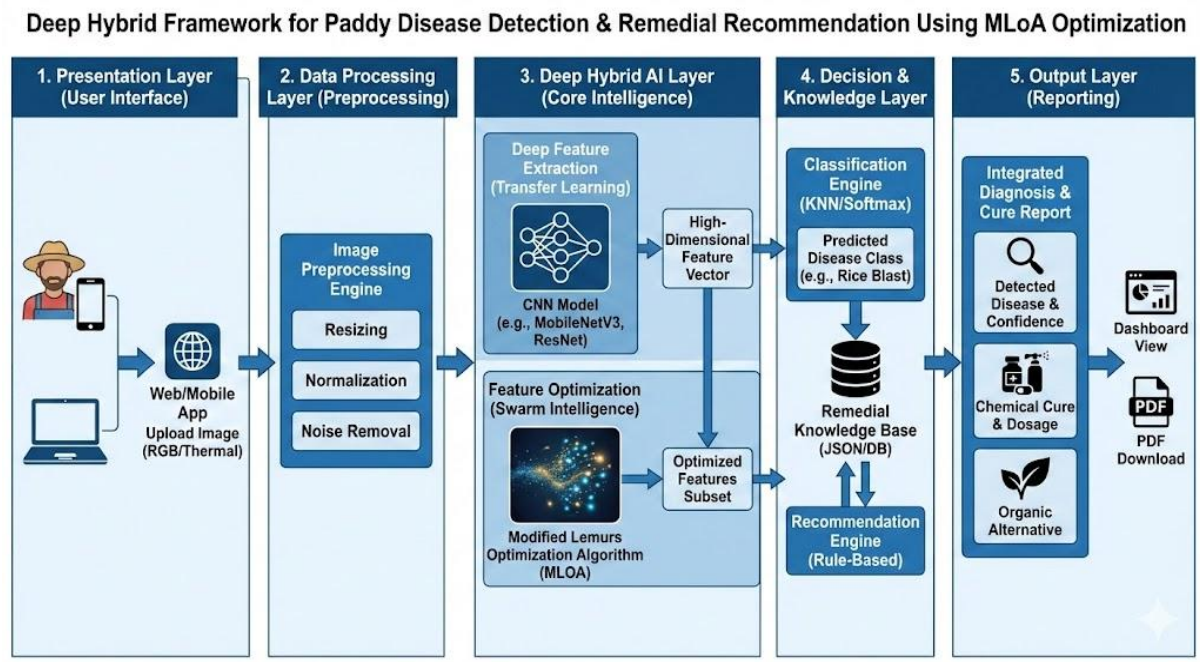
- **Deep Feature Extraction:** Instead of manual statistics, the system uses a **Transfer Learning** approach with a pre-trained CNN (e.g., **MobileNetV3**) to automatically learn and extract complex features (textures, edges, shapes) from standard **RGB images**.
- **MLOA Optimization:** The extracted deep feature vector (which may contain 1000+ features) is passed through the **Modified Lemurs Optimization Algorithm (MLOA)**. This "Swarm Intelligence" layer acts as a smart filter, selecting only the most relevant features for classification.
- **Remedial Module:** A post-classification module links the predicted disease class to a structured **Remedial Knowledge Base**, fetching specific chemical dosages and organic alternatives.
- **Web Interface:** The entire system is encapsulated in a web-based portal ("AgriCure") allowing for easy image upload and report generation.

6. Advantages of Proposed System

1. **Superior Accuracy:** By using Deep Learning features instead of simple statistics, the system can achieve accuracy levels between **95-99%**.

2. **Cost-Effective:** Works with standard smartphone images (RGB), removing the need for expensive thermal cameras.
3. **Efficiency:** The MLOA algorithm optimizes the heavy deep learning features, ensuring the classifier runs fast without losing accuracy.
4. **Farmer-Centric Solution:** Provides end-to-end support, from diagnosis ("It is Rice Blast") to treatment ("Spray Tricyclazole").
5. **Modern & Scalable:** The architecture is modular, allowing for easy updates to the disease database or the AI model without rebuilding the entire system.

7. Architecture

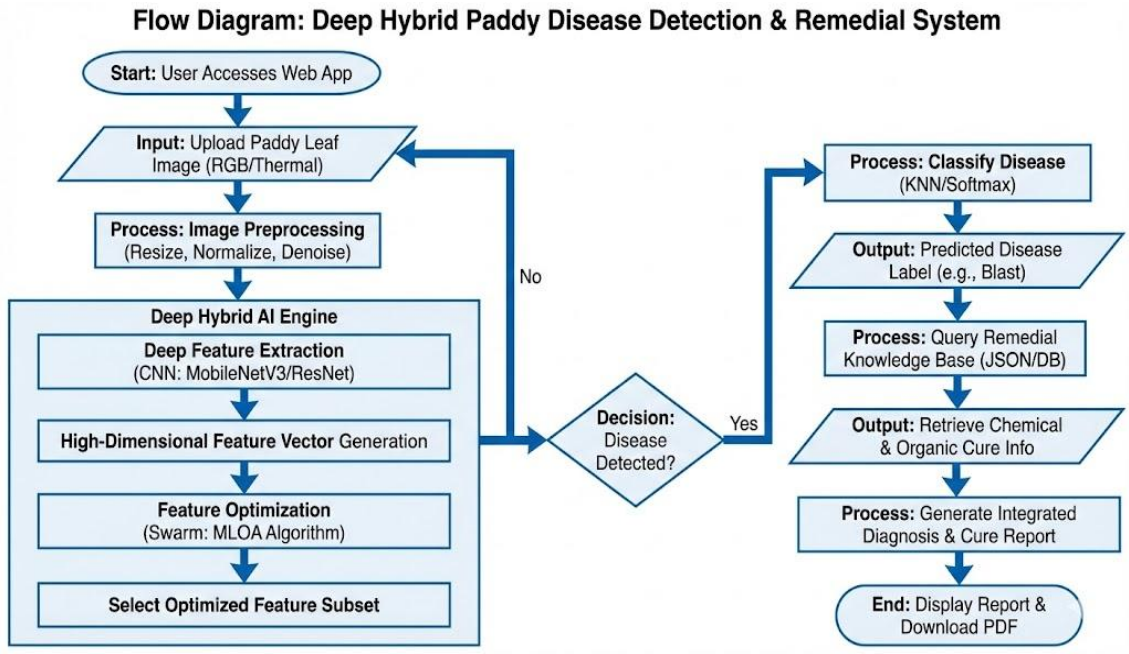


Deep Hybrid Framework for Paddy Disease Detection

- **Input Block:**
  - **Paddy Leaf Image** (RGB/Thermal)
- **Preprocessing Block:**
  - **Image Resizing** (224x224)
  - **Normalization** (Pixel scaling)
- **Feature Engineering Block:**
  - **Deep Feature Extraction:** CNN Model (MobileNetV3) $\rightarrow$ Extracts Feature Vector ($1 \times N$)

    - **Feature Optimization: MLOA Algorithm** $\rightarrow$ Selects Best Features ($1 \times K$)
- **Decision Block:**
    - **Classifier:** KNN / Softmax Layer
    - **Disease Prediction:** (e.g., Blast, Hispa, Healthy)
- **Knowledge Block:**
    - **Remedial Database:** Maps Disease $\rightarrow$ Cure (Chemical/Organic)
- **Output Block:**
    - **Final Report:** Diagnosis Name + Confidence Score + Recommended Cure

8. Flow Diagram



9. Applications

1. **Smart Agriculture:** Enables farmers to self-diagnose crop issues instantly using a smartphone.
2. **Ag-Tech IoT:** Can be integrated into drone systems for automated field surveillance.
3. **Agricultural Extension:** Assists field officers in surveying large areas and providing standardized advice.
4. **Crop Insurance:** Provides verified proof of disease damage for insurance claims.

5. **Education:** Serves as a training tool for botany and agriculture students.

10. Modules

1. System Modules (Backend)

- **1.1 Data Collection:** Gathering and organizing the **Paddy Doctor** (RGB) and **Kaggle Thermal** datasets.
- **1.2 Data Preprocessing:** Algorithms for noise removal, resizing, and augmentation to prepare images for the AI model.
- **1.3 Deep Feature Extraction:** Implementation of the **MobileNetV3** CNN architecture to extract high-level visual features.
- **1.4 MLOA Optimization Engine:** The core algorithm that processes feature vectors to select the optimal subset, reducing dimensionality.
- **1.5 Remedial Logic:** A lookup system that retrieves treatment data based on the classification result.
- **1.6 Output Generation:** Formats the analysis into a readable PDF or web report.

2. User Modules (Frontend)

- **2.1 Upload Interface:** Allows users to select images from their device.
- **2.2 Diagnosis View:** Displays the detected disease and the confidence percentage.
- **2.3 Treatment Card:** Shows the specific medicine, dosage, and safety instructions.
- **2.4 History Log:** (Optional) Saves previous scans for the user's reference.